The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction: The Unified 11-Part Search-and-Rescue Plot

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

We formalize the system trajectory of an asset as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

3 The 27-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 27}$. For each node i, the vector decomposes into three distinct socio-pragmatic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{empirical}} \mid \boldsymbol{c}_i^{\text{cognitive}} \mid \boldsymbol{s}_i^{\text{institutional}} \right] \tag{2}$$

where $\boldsymbol{e}_i \in \mathbb{R}^{18}$ encapsulates the Barna loop splits, the 72% Pew United States Church macro-advection wavelets, and the JAX spatial grid coordinates. The cognitive component $\boldsymbol{c}_i \in \mathbb{R}^6$ packs the compiled LISP bar preferences and the 3-dimensional Rational Speech Act (RSA) scalar implicature distributions. The institutional component $\boldsymbol{s}_i \in \mathbb{R}^3$ isolates the zero-uncertainty Searlean speech-act flags (X counts as Y in C), allowing the network to collapse giant graph topologies down to a unified virtual 0-simplex token ready for decision tree lookup routing.

4 Time-Bounded Deontic Logic and Legal Contract Invariants

We formalize commercial billing compliance as a temporal trajectory tracking process within a bounded legal lattice, where obligations switch from voluntary commitments to enforced penalties as a function of discrete time steps t.

4.1 The Temporal Utility Decay Function

Let $\mathcal{C}_{\text{bill}}$ be the brute invoice cost, $\mathcal{P}_{\text{base}}$ be the initial late penalty modifier, and t_{deadline} represent the explicit legal temporal boundary. The agent's choice space maps to $\mathcal{A} \in \{\text{Pay}, \text{Evade}\}$. The utility of compliance remains constant, while the utility of evasion scales non-linearly post-deadline via a directional advection step:

$$\mathcal{U}(a = \text{Pay}, t) = -\mathcal{C}_{\text{bill}} \tag{3}$$

$$\mathcal{U}(a = \text{Evade}, t) = \begin{cases} 0, & \text{if } t \leq t_{\text{deadline}} \\ -\mathcal{P}_{\text{base}} \cdot \gamma^{(t - t_{\text{deadline}})}, & \text{if } t > t_{\text{deadline}} \end{cases} \tag{4}$$

where $\gamma > 1$ represents the compounding interest escalation parameter.

4.2 Softmax Choice Vector Convergence

The probability distribution governing contract compliance is compiled via the exponentiated softmax calculus ($wpp = \exp(\beta \cdot \mathcal{U})$):

$$P(a = \text{Pay} \mid t) = \frac{\exp\left(\beta \cdot \mathcal{U}(\text{Pay}, t)\right)}{\exp\left(\beta \cdot \mathcal{U}(\text{Pay}, t)\right) + \exp\left(\beta \cdot \mathcal{U}(\text{Evade}, t)\right)} \tag{5}$$

When $t > t_{\text{deadline}}$, the non-linear penalty block forces an immediate collapse of the choice matrix, driving $\lim_{t \to \infty} P(a = \text{Pay} \mid t) = 1.0000$. This mathematically establishes the transition from a Commissive intent to a Directive-enforced institutional reality, unblocking Mesarovic's Attainable Space V' for automated legal compliance auditing.